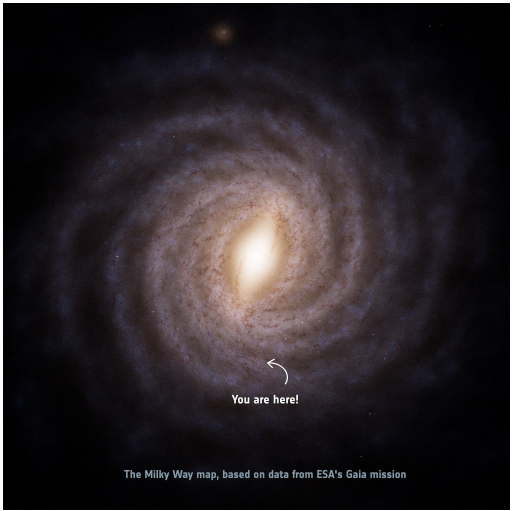
Gaia

Measuring the positions of two billion stars

SRMP 2026

Session 1

A map of our galaxy



ESA, CC BY-SA 3.0 IGO

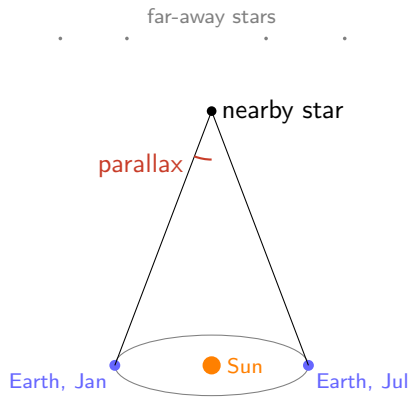
- We live inside the Milky Way, so we can't step outside and take a picture of it.
- Instead: measure where each star is and how it moves, then build the map.
- Gaia measured positions for almost 2 billion stars, and distances and motions for 1.5 billion. That's about 1% of the stars in our galaxy.
- This picture is not a photo: it's an artist's drawing, based on Gaia's measurements.

How far away is a star? Parallax

- Hold up a finger and blink one eye, then the other. Your finger jumps.
- Same idea: look at a star in January, then in July. A nearby star seems to shift against the far-away ones.
- The closer the star, the bigger the shift.

$$\text{distance [pc]} = \frac{1}{\text{parallax [arcsec]}}$$

1 parsec (pc) = 3.26 light-years. An arcsec is a tiny angle (next slide).
Nearest star: parallax 0.77 arcsec, so distance $= 1/0.77 = 1.3$ pc.



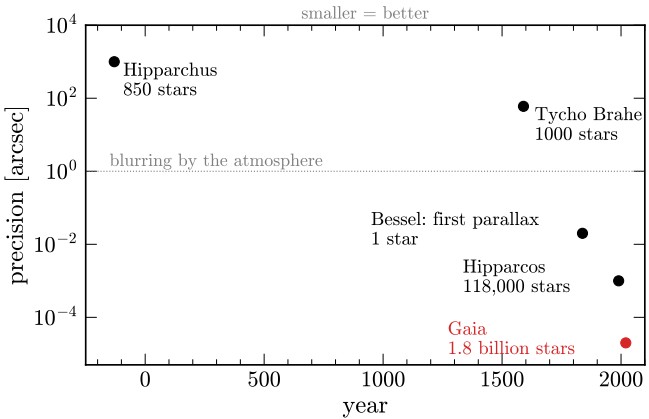
These angles are tiny

1 arcsec (arcsecond) = 1/3600 of a degree.

	angle (arcsec)
The full Moon	1800
Smallest detail your eye can see	60
Parallax of the nearest star	0.77
Gaia's precision, bright stars	0.00002

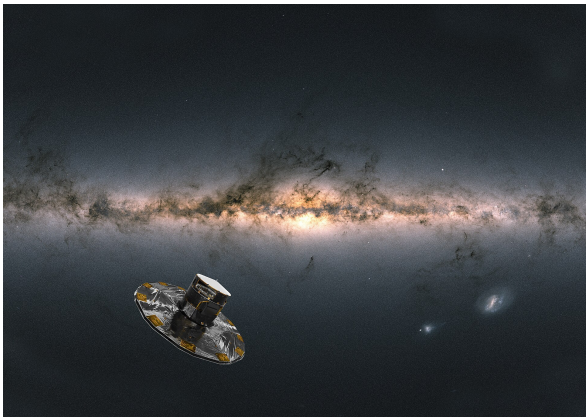
- 0.00002 arcsec is the width of a human hair seen from 1000 km away (about Boston to Detroit).
- Astronomers often use milliarcseconds: 1 mas = 0.001 arcsec. So Gaia's best is 0.02 mas.

2000 years of measuring stars



- Each step down the left axis is 100 times smaller.
- The atmosphere blurs stars to about 1 arcsec. Careful work from the ground beats that a little; space does far better.
- Gaia: 50 times more precise than Hipparcos, for 15,000 times as many stars.

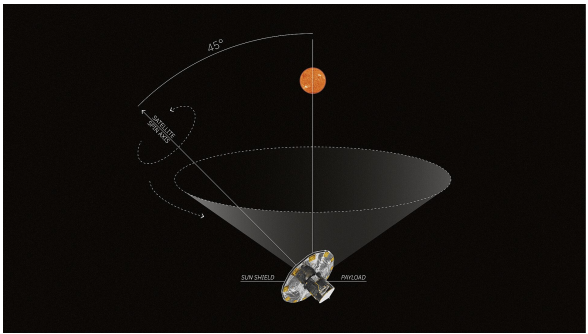
The spacecraft



ESA, CC BY-SA 3.0 IGO

- Launched by the European Space Agency (ESA) in December 2013.
- Sat 1.5 million km from Earth (4 times farther than the Moon), with the Sun, Earth and Moon all behind it.
- Two telescopes, one giant camera: almost 1 billion pixels.
- The 10 m sunshield blocks the Sun, so the telescopes stay cold and steady.
- Observed for 10.5 years, until January 2025.

How Gaia scanned the sky



Space Travel Blog / R. M. Jansone / M.-L. Aru, CC BY 4.0

- Gaia spun slowly, once every 6 hours, sweeping its telescopes around a big circle on the sky.
- The direction it spun around also slowly changed, so over months the circles covered the whole sky.
- Over 10 years, each star was seen about 140 times.

Watch: Gaia scanning the sky
youtube.com/watch?v=BnFyzZGWuYs

What Gaia measures



Where

Position on the sky, again and again.

→ distance (parallax)
→ how fast it moves across the sky



Brightness and colour

Through three filters: one wide, one blue, one red.

→ how hot the star is
→ whether it changes brightness



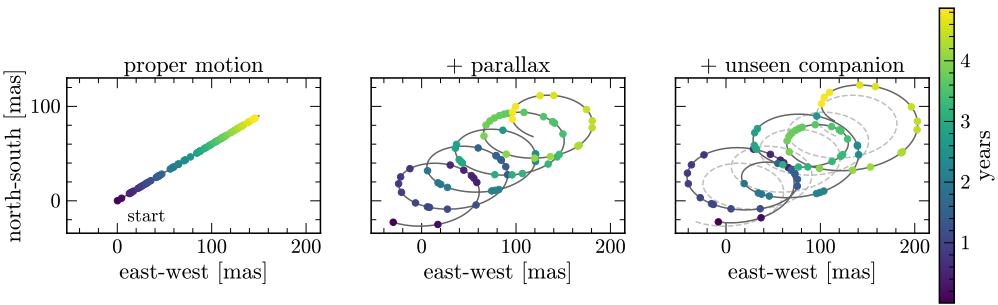
Speed toward or away

By splitting its light into a rainbow, or spectrum (bright stars only).

→ the third direction of motion

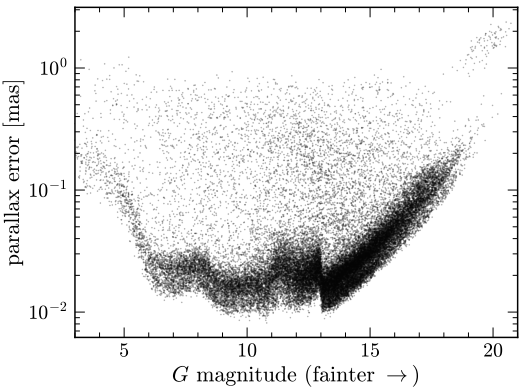
Together: a 3D map of the stars nearby, and how they move.

The path of a star across the sky



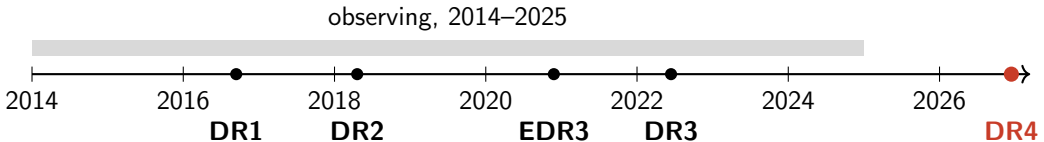
A pretend star, seen 70 times over 5 years (colour = time). Left: it drifts in a straight line. Middle: Earth's yearly trip around the Sun adds a loop (parallax). Right: an unseen companion adds a wobble on top. The dashed line is the path without it. (The wobble is exaggerated.)

How precise is it?



- Each dot is a star within 50 pc.
- G magnitude: how bright a star looks. It runs backwards: bigger number = fainter.
- Fainter stars send less light, so their parallax is less certain.
- The step at $G \approx 13$ comes from how the camera handles bright stars.

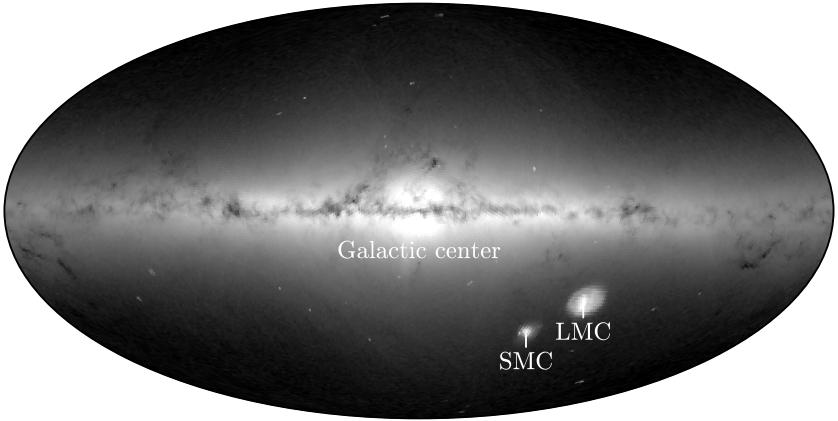
Data releases



DR = data release. EDR3 was the first half of DR3.

- **DR3** (2022): what we'll use in the tutorials. Average position, distance and motion for each star.
- **DR4** (2 December 2026): 5.5 years of data, and every single measurement of every star, not just averages.
- **DR5** (around 2030): all 10.5 years.

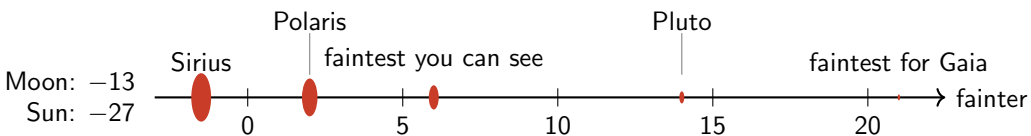
The whole sky, as seen by Gaia



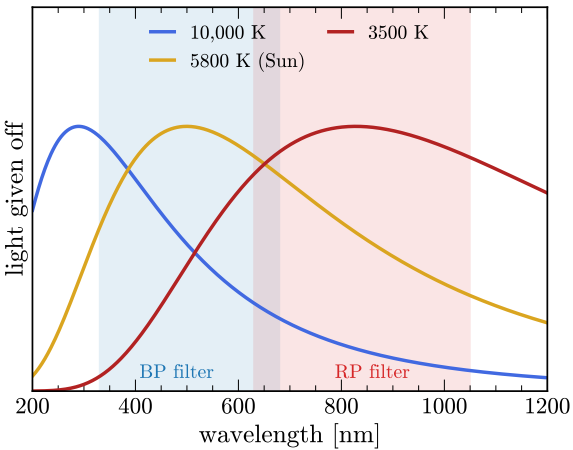
Number of stars per patch of sky in DR3. The whole sky is flattened into an oval, like a world map. The Milky Way is a flat disk and we sit inside it, so it looks like a band across the middle. Dark lanes are dust clouds. LMC and SMC are two small galaxies orbiting ours.

How bright does a star look? Magnitudes

- Gaia's camera counts how much light arrives from each star.
- Astronomers measure brightness in magnitudes. The scale runs backwards: bigger number = fainter.
- 5 magnitudes = 100 times fainter.
- Gaia's main brightness, G, is measured through its wide filter.

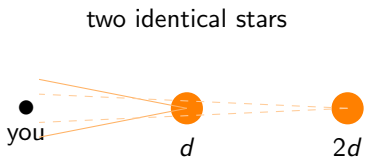


What colour is it? Temperature



- Hot stars glow blue-white, cool stars glow red.
- Gaia measures each star through a blue filter (BP) and a red filter (RP).
- Colour $= BP - RP$. Small: hot. Big: cool. The Sun: 0.8.
- Star types, hottest to coolest: **O B A F G K M**. The Sun is a G star.

How bright is it really? Absolute magnitude



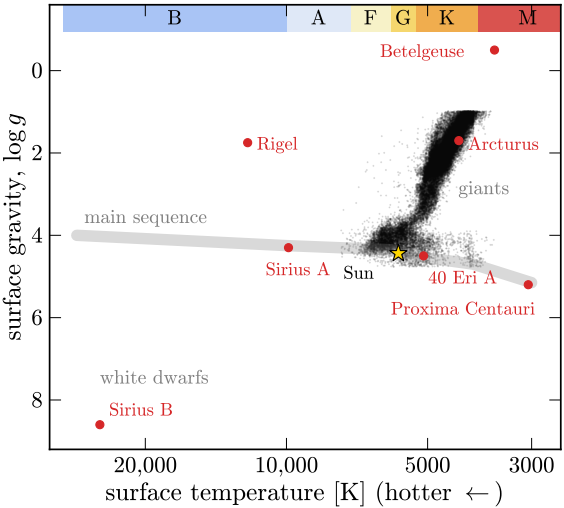
Twice as far away: 4 times fainter.

- A star can look faint because it gives off little light, or because it's far away.
- Absolute magnitude: how bright a star would look from 10 pc.
- With Gaia's parallax (in mas):

$$M_G = G + 5 \log_{10}\left(\frac{\text{parallax}}{100}\right)$$

- 40 Eri A: $G = 4.18$, parallax 199.6 mas, so $M_G = 4.18 + 1.50 = 5.68$. The Sun's is 4.67: 40 Eri A gives off less light.

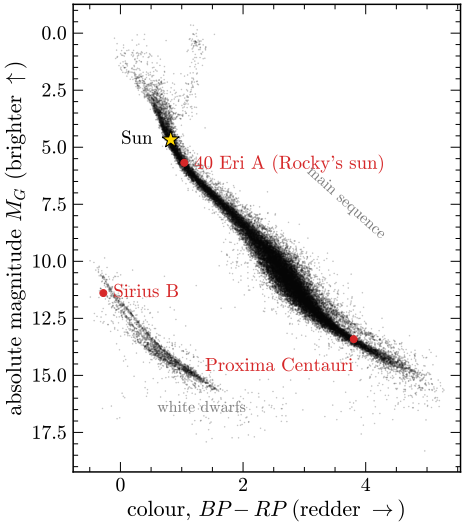
Temperature and gravity



Black dots: 50,000 stars measured with Gaia's spectrograph.

- A star's spectrum (its light split into a rainbow) gives its temperature and the gravity at its surface.
- Big, puffy stars: weak gravity. Small, dense stars: strong gravity.
- $\log g$: each step of 1 is 10 times stronger. Sun: 4.4. Earth: 3.0.
- **Main sequence:** stars like the Sun.
- **Giants:** old stars that have swollen up. Betelgeuse in place of the Sun would reach past Mars.
- **White dwarfs:** the size of Earth, as heavy as the Sun.

What Gaia sees: stars within 50 pc



The Hertzsprung–Russell (HR) diagram

- Gaia has no detailed spectrum for most stars, but it has colour and distance for over a billion.
- Colour tells us temperature. Absolute magnitude tells us how much light the star gives off.
- Same groups as before: the main sequence, white dwarfs at the bottom left, and a few giants at the top. Giants are rare this close to us.
- In *Project Hail Mary*, Rocky's sun is a real star: 40 Eri A, 5 pc away.

Your turn: real Gaia data

You'll download Gaia data yourself and remake the plots from today.

```
SELECT source_id, ra, dec, parallax, phot_g_mean_mag, bp_rp
FROM gaiadr3.gaia_source
WHERE parallax > 20
```

- In words: "For every star with parallax above 20 mas (closer than 50 pc), give me its ID, position on the sky (ra, dec), parallax, brightness and colour."
- The Gaia archive: https://gea.esac.esa.int/archive/

References and image credits

- Gaia Collaboration, Prusti et al. 2016, *The Gaia mission*, A&A 595, A1
- Gaia Collaboration, Vallenari et al. 2023, *Gaia DR3: Summary of the content and survey properties*, A&A 674, A1
- Lindegren et al. 2021, *Gaia EDR3: The astrometric solution*, A&A 649, A2
- Recio-Blanco et al. 2023, *Gaia DR3: Stellar parameters from RVS spectra*, A&A 674, A29
- ESA Gaia pages: https://www.cosmos.esa.int/web/gaia

Images (via Wikimedia Commons):

- Milky Way map: ESA, CC BY-SA 3.0 IGO
- Gaia spacecraft: ESA, CC BY-SA 3.0 IGO
- Gaia scanning diagram: Space Travel Blog / R. M. Jansone / M.-L. Aru, CC BY 4.0

Sky map, HR diagram and other plots made from Gaia DR3 data with the tutorial notebooks.